A Topology of Variables Impacting Voting Behaviour in the 2024 United Kingdom General Election

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Abstract

Background

In light of economic events and migration shifts since the 2019 election and post-Brexit, electoral trends have shifted as a result of changes in political and social compositions within the UK population. Labour winning the recent GE24 has provided stronger rationale to employ an approach that observes the interplay of these dynamic forces. Yet, the majority of past studies have used traditional statistical approaches which, although insightful in comprehending theoretical frameworks to voting behaviour, are limited in their linearity. This dissertation aims to investigate how socioeconomic and demographic variables as joint distributions may affect voting decisions in the UK in the recent election, especially in relation to migration trends at constituency-level.

Methodology

Topological Data Analysis BallMapper (TDABM) was used in accompaniment to Ordinary Least Squares (OLS) to explore multidimensional variables in the 2024 UK General Election. Using merged data from Census 2021 and the 2024 GE results at constituency-level, TDABM supplements visualisations through observing joint distributions of socioeconomic, migration and demographic data through interconnected balls.

Results

This study suggested that voters' socioeconomic, migration, and demographic status do impact voting behaviour particularly at constituency-level, revealing the geographical heterogeneity of Labour winning areas and homogeneity in Conservative strongholds.

Conclusion

Findings of this study on the heterogeneity of variables between parties make practical contributions to electorates/political parties regardless of political orientation (left-right) to aim to bridge the urban-rural divide as joint distributions of varying factors are revealed. Indeed, proposing manifestos or policies reflective of the geographical socioeconomic and demographic state prior to upcoming elections could lead to a trajectory of the nation's economic prosperity from more homogenous behaviours - in light of political values.

Acknowledgements

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Declaration

No portion of the work referred to in this extended research project report has been submitted in support of an application for another degree or qualification of this or any other university or other institute of learning.

Intellectual Property Statement

1. Introduction

The 2024 United Kingdom General Election (UK GE) was called by former British Prime Minister Rishi Sunak on July 4, following a Conservative reign since 2010. In this election, Labour won by a landslide majority[1]. Many social and economic influences shifted the political landscape leading up to the election – mainly exogenous and macro-level (Chrysantou & Guilló, 2024). Theoretically, traditional frameworks like Pocketbook Voting (Lewis-Beck, 1985) and Rational Choice Theory (Fedderson, 2004) have provided strong foundations on the roles these factors play on voting behaviour, emphasising socioeconomic/demographic impacts like employment status and educational attainment. Under the Conservative rule, these were events such as unemployment, migration, Brexit, and COVID-19 (Portes, 2023).

Nevertheless, theories offering insights into voting behaviour cannot fully address the complexity and multi-dimensionality of significant variables, such as the intersectionality between a voter's demographic profile and their socioeconomic state, making advanced social analytical approaches necessary. Hence, advancements unraveling these impacts have recently grown in light of economic instability because, if the election serves as a medium for political accountability, there is an expectation for voters to support/penalise incumbent parties from their perceived outcomes (Powell, 2000; Hobolt, 2016; Matakos and Xefteris, 2020). For political parties to formulate policies aligned with impactful outcomes reflective of diverse voter groups across areas, employing a more sophisticated technique which explores high-dimensional multivariate interactions is imperative for a better democracy.

[1] Labour secured 412/650 seats. House of Commons Library (2024). *General Election 2024 results*. Available at: [https://researchbriefings.files.parliament.uk/documents/CBP-10009/CBP-10009.pdf]

Migration and economic patterns post-Brexit has further increased the complexity of the UK's electoral environment (Portes, 2023). For instance, immigration/emmigration of individuals due to policies that may advantage/disadvantage them can shift demographic constituents, consequently altering voting choice. This underlines a geographical-focused analysis to capture these intricacies. Indeed, the reliance of traditional statistical approaches in showing interactive relationships to reflect theoretical frameworks justifies the need for a more innovative approach, as whilst conventional methods strengthen framework foundations, they are often limited by dimensionality and linearity (Rudkin et al., 2023). With UK voting datasets being large, an approach that grasps high-dimensionality to identify underlying relationships is important – one such as Topological Data Analysis BallMapper (TDABM) (Dłotko, 2019). TDABM mitigates oversimplification, allowing for cluster identification in multiple axes. In this context, traditional methods complemented by TDABM through its joint distributions offer much comprehension on society's voting behaviours, which the former might overlook.

This study aims to employ TDA to explore the impacts of socioeconomic, migration, and demographic profiles on the 2024 GE at constituency-level, addressing three research questions – whether changes to these variables (e.g. employment/migration status) predict shifts in voting behaviour, the extent of its predictability, and its implications on the UK political landscape. Using 2021 Census data, the analysis covers variables like voters' income, employment status, education, migration status, age, and household composition.

By integrating these variables, this study approach contributes to the growing field of TDA in electoral analyses and social sciences in general. Furthermore, this study adds into the current extensive understanding on significant variables impacting voting behaviour, but with an additional lens of how factors interplay in the current sociopolitical climate, as this enhances voting prediction reliability based on available data. Indeed, a deeper understanding of this offers more inclusive policies that would subsequently alter electorates' voting decisions more aligned with the UK's democratic system.

2. Literature Review

2.1 Context: The 2024 UK General Election & Voting Influences

Prior to the 2024 Election day, polls had already predicted a Labour win, and the party's subsequent victory after over a decade had sparked curiosity on the potential drivers to this historical change[2]. Current literature highlights socioeconomic, migration, and demographic factors as key attributes among many others (Chrysantou & Guilló, 2023; Anderson, 2007), and theoretical frameworks which help comprehend voting behaviours have provided strong foundations in the roles of these factors. For instance, "pocketbook voting" posits that voters elect based on their economic self-interest– incumbent party support would continue if they are financially stable and vice-versa (Lewis-Beck, 1985). Rational Choice Theory (Fedderson, 2004) in voting also suggests voters choose based on their "investment benefit" and social consequences. As empirical findings further suggest that voters would typically punish/reward ruling parties based on their economic outcomes (Duch & Stevenson, 2008; Lewis-Beck & Stegmaier, 2000; Van Der Brug et al., 2007), it is thus inferential to suggest that after a period of economic distress, the incumbent government would be voted out. This is evident in the recent economic instability in the UK under Conservative rule due to crises like the cost of living, COVID-19 and Brexit, as there were significantly higher rates of economic dissatisfaction towards the government in 2024 following a steady increase in every election year since 2015[3]. Hobolt and Tilley's (2016) empirical study reinforced these findings, further adding on the inclination for the economically disadvantaged to turn to

[2] Voting polls were commonly discussed by major platforms like *YouGov, Statista, Kantar,* showing Labour wins, compiled by *BBC* (2024). Available at: [https://www.bbc.com/news/uk-politics-68079726]

[3] British Election Survey (2024). Available at [https://www.britishelectionstudy.com/2024-general-election/bes-insights-incumbency-and-dissatisfaction-in-the-2024-uk-general-election/]

far-right parties. In the recent election, this was observed with Nigel Farage's radical Reform UK party winning seats from the Tories who dropped to its lowest seats since 1998[4].

Labour's win also reopens discussions on the "Red Wall" collapse and the geographical-based concept of "left-behindedness" to explain the phenomenon of political realignment (English, 2022; Dijkstra et al., 2020). Although many constituencies gradually swung to Conservative up to 2019, this trajectory changed in favour of Labour who regained its seats in areas previously lost in 2024, raising questions on how far this perceived hypothesis holds. Indeed, there are diverse voter profiles across rural/urban constituencies that these labels fail to capture - even Fieldhouse & Bailey (2023) argued that Labour "lost votes just about equally everywhere".

Therefore, understanding the complexities of voting attitudes is crucial for several reasons. Firstly, it helps decipher the ongoing debate on the UK's political realignment/dealignment. It also influences future policies aiming to strategically address the nation's significant underlying socioeconomic problems, which may serve either as a risk or opportunity for political parties in their incumbent constituencies.

2.2 Prosperity and Politics: Socioeconomic Influences on Voting Behaviour

Under Pocketbook voting, exploring the validity of this hypothesis requires an evaluation of the interactions between individual circumstances with other socioeconomic factors, such as income levels, education and employment. Kulachai et al. (2023) in their review provided sound evidence on how income levels significantly predict voting behaviour

[4] The Guardian (2024). Available at: [https://www.theguardian.com/politics/ng-interactive/2024/may/03/local-elections-key-charts-showing-tories-slump-to-worst-performance-since-1998]

- the higher the individual's income, the more right-leaning policies they would support. Conversely, those more deprived would support policies which prioritise social welfare. However, Bartels (2016) added another important layer of complexity suggesting that unequal income gaps are a significant predictor to voting choice, as voters would support parties that introduce policies mitigating economic disparity. This discredits the linear relationship between income and electoral decision, as income interacts with factors like employment and education level through regional disparities. For example, regardless of income, Kulachai et al. (2023)'s review added that those with higher education levels would rank policy issues differently than those of lower education levels.

This brings to another socioeconomic variable pertinent to voting decisions - education. Indeed, this has been a well-established finding, prominently from Nie et al. (1979)'s study suggesting education as one of the strongest predictors to electoral choice, and in fact, the strongest for Leave-Brexit (Simon et al., 2024). Individuals with higher educational attainment are more likely to vote for progressive and left-leaning parties (Evans & Anderson, 2006; Van der Brug et al., 2007; Kulachai et al., 2023). With higher political awareness and participation, those with higher educational attainment tend to influence the political landscape more when engaged as they have better comprehension of the country's societal challenges (Kulachai et al., 2023).

Nevertheless, authors should not confuse a highly educated individual's political influence with their political compass, as there seems to be multifaceted relationships between this and voting behaviour. For example, in the UK, the education divide for Labour-Conservative votes was not stark until post-Brexit, where the pattern of graduates increasingly voting Labour was significantly noticed for the first time (Ford et al., 2021).

While there has been seminal studies to observe this pattern, there also seems to be regional variations to voting preferences limited in exploration. This was pointed out by Simon et al. (2024) observing the geographical heterogeneity of political choice even among those of similar qualifications. It is therefore sound to suggest that there is a spatial heterogeneity to the educational effect, making constituency-level observations important.

It goes without saying that individuals use educational qualifications to secure stable income through employment (Capsada-Munsech, 2017). Thus, status of one's employment is another critical factor to voting behaviour, with those precariously employed being more likely to select parties that address employment security and economic mitigations (Turner, 2022). This explained the Conservative government remaining in office, especially during the Coalition government period, as there were policies specifically inplace to mitigate unemployment in the wake of the 2008 financial crisis (McKnight & Cooper, 2022). Despite this factor, if economic conditions are unexpectedly bad and unemployment rates rise, voter dissatisfaction could lead to a reform and change in party support (Lewis-Beck & Stegmaier, 2000) - aligning with pocketbook voting theory. With the rise of UK unemployment rates under the Tories post-pandemic, this has been evident with the nation voicing out concerns unaddressed leading up to government change, highlighting how the nation's employment state can evoke change in the political landscape. Additionally, not all constituencies equally flourish with globalisation, leading up to "left-behind" areas constituting voters known to work routine jobs and causing the rise in anti-immigration sentiments for "stealing jobs", typically voting Conservative/Leave (Jennings & Stoker, 2017).

2.3 Migration and Voting Pre-Brexit

Even before the referendum, immigration and the economy were the two highest pressing issues to GE15 voters (Ipsos-MORI, 2022; Figure 1). This was not uncommon as these have consistently been the nation's annual political concern since 2002 (Ford and Morris, 2022).

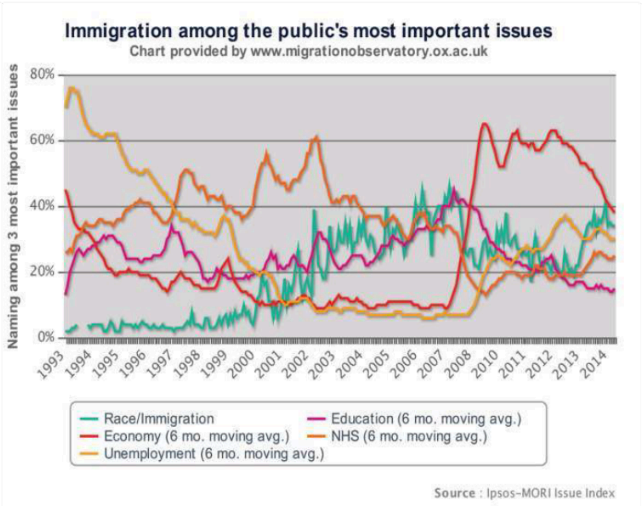


Figure 1: Most Important Issues in the UK 1993-2014 (Ipsos-MORI, 2022)

The EU imposing pro-immigration policies in 2016 was perceived to restrict the nation's autonomy, and with voters' preference for tighter immigration restrictions due to economic concerns, the Brexit referendum was proposed. The "Leave" campaign resulting from this rooting for immigration control shifted voting patterns, mainly promoting the Conservative government to win the 2017/2019 elections (Portes, 2024). Anti-immigration sentiments particularly appealed to far-right politics like UKIP (Ford and Goodwin, 2014), as McManus (2023) highlighted migration as an important factor that catalysed right-wing party electoral gains in certain countries in the past, making Brexit a significant landmark in voting compositions.

2.4 Migration Trends Post-Brexit

Since Brexit, immigration policy changes were introduced by the incumbent government to recover economically, encouraging higher migration rates. For example in 2021, the introduction of the points-based immigration system, less stringent job restrictions in social/healthcare sectors due to shortages post-COVID-19 and humanitarian aid visas for Ukrainians/Hong Kongese[5]. With the influx of net-migration from 100,000 to 700,000 since the pandemic (ONS, 2024)[6], it is no surprise that the demographic makeup of constituencies will have strikingly changed, directly influencing the socioeconomic state of UK regions, and consequently its electoral patterns. While some areas might benefit from migrants' economic gains, others might experience tensions as public concerns have been raised.

Since the Red Wall downfall in GE19 and the Tories introducing more stringent immigration policies by the end of 2023 (Manning, 2023), migration shifts may disrupt typical voting compositions. For example, in GE24 Labour regaining lost seats of past elections could be a testament to political realignment for constituencies with higher migration rates. Whether true or not, investigating the area-based extent to which migration affects voting behaviour is crucial especially with migration and the economy being top concerns, as this sets the political scene preceding/following elections.

[5] UK Home Office (2024). Available at: [www.gov.uk/government/statistics/immigration-system-statistics-year-ending-march-2024]
[6] Office for National Statistics (2024). Available atL [https://www.ons.gov.uk/peoplepopulationandcommunity/populationandmigration/internationalmigration/articles/internationalmigrationresearchprogressupdate/may2024]

2.5 Demographic Dynamics: A Closer Look on Electoral Outcomes

In commensurate with increased net migration, population shifts in constituencies reflectively become drastic. This triggered the call for the recent electoral boundary change in 2023 to ensure fair voter representation (Fieldhouse & Bailey, 2023). The new demographic nuances among voters play a pivotal role in understanding voting predispositions, evident in the thorough studies on voter demographics by researchers in the past (e.g. English, 2022; Fieldhouse & Bailey, 2023). This introduced concepts like “left-behindedness” to help understand sociodemographic profiles in less progressive areas (Jenning & Stoker, 2017). However, it can be argued that this reductive label generalises diverse populations, leaving out the opportunity to further understand different characteristics in a population/area which predispose them to a party.

For instance, studies have generally observed more varying voting patterns in younger and diverse groups than older, like groups (Kulachai, 2023). In terms of immigration, older and less educated populations in local areas with increasing immigration rates tended to elect parties with anti-immigration agendas (e.g. Conservative, Reform UK), regardless of whether proposed policies were successful or not (Hampshire, 2015). Bigger family households may also be indicative of underlying social dynamics, with voters more likely rooting for improved social welfare (Wolfinger & Wolfinger, 2008). Although current findings have deduced these direct relationships, imperative to these factors is the geographical context which may account for finer differences. While Johnston & Pattie (2000) established the term “neighbourhood effect” in voter behaviour where “people who walk together vote together”, Horan et al. (2024) more recently highlighted that election results are not necessarily similar within constituencies of similar demographics but of different areas, challenging the credibility of place-based concepts like ‘left-behindedness’. This is further reinforced by Dorling (2016) who suggested that populations have been more geographically segregated in the UK now since the 70s, calling for a more careful approach in observing regional differences – especially with recent dynamic migration trends changing constituency boundaries and demographics in 2023.

2.6 Current Approaches

While empirical and theoretical findings in political literature support swings observed in the recent elections in terms of understanding party polarisation, they do not necessarily consider how major variables contributing to voting shifts interplay within particular geographies. For instance, Chrysanthou & Guilló (2023) and Evans & Andersen (2006) posited that individual socioeconomic perceptions which influence their voting choices in an area can be determined through personal attributes and demographics, yet these are seldom analysed together. If joint distributions help understand voters’ political compass in an election, reduced statistical models are simply insufficient to infer variables impacting voting behaviours – apparent in Anscombe’s Quartet (1973), where similar regression coefficients are not necessarily generated from data with similar distributions. This underscores the need for an appropriate approach to visualising data when observing underlying patterns.

Regression models commonly utilised in this expansive literature have been useful in capturing the relevance of significant variables in voting behaviour (see Chrysanthou & Guilló, 2018, Johnston & Pattie, 2004), but its linear statistical approach raises concerns over the endogeneity of voting preferences. With extensive studies pointing out regional heterogeneity in factors like education, employment and age on electoral gains, it is just to underline the crucial role that geographical contexts play, yet surprisingly this level of granularity is often ignored. Despite the growing literature on using demographic data to understand voting patterns through various methodologies, findings on neighbouring constituency nuances for the UK GE remain limited. Indeed, the Brexit referendum is a compelling example on why understanding the complexity of variables is important on voting behaviour. The Migration Observatory (2020) highlighted the discrepancies amongst areas with differing levels of immigration and the influence on the Brexit vote, which gives consideration to the dynamic interplay between voting and regional level influences. Rudkin et al. (2023) in their TDA study provided empirical evidence on other economic and sociodemographic-related variables with the Brexit vote at constituency-level. Their finding of homogeneity in the Leave vote in certain regions is one where standard statistical methods are unable to capture.

This sets a strong motivation for a more thorough approach in analysis, one that does not obscure nuances and captures geographical contexts – TDABM. Initially established in the social sciences, there remains a lacuna of research employing TDA to understand voting patterns despite its advantages of tackling statistical limitations. Thus, not only does the employment of this method contribute to the nascent field of TDA in data science in the social sciences, it also enables comparisons with other existing methodologies like standard regression models.

TDABM is a powerful approach to mitigate the methodological, empirical and theoretical limitations as it visualises the joint distributions (Dlotko, 2019), linking voter socioeconomic characteristics and their political leanings rather than viewed separately.

Therefore, it allows viewing the shape of major party support (Labour/Conservative) across the demographic space on top of unravelling multi-directional interactions between variables and voting behaviour. Considering the large UK voter population, TDABM is also useful in handling multidimensional and heterogenous data such as Census. TDABM enhances current common approaches by uncovering intricate structures/patterns across constituencies where traditional methods might not pick up. With recent boundary changes, TDABM allows for the exploration of spatial data and the delineation of areas formerly grouped similarly, giving it all the more reasons to delve deeper into analysing these subtle distinctions at constituency-level.

Since the last GE, many economic and policy changes have occurred, and the extent to which these multiple factors may have influenced the political landscape in the GE24 remains unknown. By comparing to traditional statistical approaches like regression, this study aims to understand how local socioeconomic, demographic and migration variables influence voting behaviours at constituency-level in the recent 2024 UK GE using Topological Data Analysis, and the extent to which this can be inferred to voting behaviours in the past and in future elections. These are addressed through exploring three main research questions which are:

1. How do socioeconomic variables predict constituency-level voting patterns in the UK?
2. To what extent do migration patterns post-Brexit impact the political landscape in the UK at constituency-level?
3. Can demographic changes (e.g. age distribution and household composition) predict shifts in voting behaviour from migration trends?

3. Methodology

3.1 Topological Data Analysis BallMapper (TDABM)

Integral to this methodology is the use of TDA BallMapper (Dlotko, 2019). Due to the nature of the multidimensional Census data, TDABM is appropriate to infer any correlations, done by representing data as point clouds of interconnected balls in a TDABM graph to observe its data shape. In the graph, each node/ball depicts constituencies sharing similar characteristics, its size representing the proportion the constituency comprises, and the colour intensity reflects on the behaviours of relevant variables.

As observed in current literature, researchers tend to view variables impacting voting behaviour through its subsets and categories due to the inevitable multicollinearity when it comes to aggregated demographic spaces (Rudkin et al., 2023). TDABM alleviates this limitation by joining distributions of all relevant variables whilst not depending on aggregated visualisations, thereby providing more geographical nuance. This is clear in the fact that Rudkin et al. (2024) in their TDA approach to the Brexit vote revealed more homogeneity of the Leave vote in certain areas than the contrary.

In the context of this particular study where the social science domains of geography and politics in data science intersect, TDABM therefore becomes a suitable approach as it consists of topological features (which reveal the shape of data), whilst providing implications on these features' relations (through balls/clusters) (Otway & Rudkin, 2024). Furthermore, the arrangement of each node in the topological space enables visualising the distribution of party compositions at constituency-level across variations of socioeconomic/demographic variables. To observe any electoral shifts, the analysis will investigate any emerging clusters. TDABM does not show the significance level of each feature in these clusters, however this can be mitigated by complementing it with regression models which will also be applied in this study prior to TDA.

3.2 How TDABM Works

For simple understanding of the BallMapper algorithm, the BM functions on a point cloud *X*. For each dataset, the algorithm first selects a random point *xi* and covers a "ball" *X* of a certain radius around it. This only parameter is ε – a positive constant that determines the ball radius in the TDABM graph. The radius of balls impacts the intricacy of the BM analysis, where the smaller the radius, the more granular the picture. Nevertheless, a radius too small may disrupt the bigger context and vice versa, both risking the revelation of important insights. This highlights the need to choose a suitable trade-off.

Next, the BM randomly and repetitively selects other uncovered points, also called landmarks, until all landmarks within ε are covered. The algorithm will then illustrate a graph where landmarks *B(X)* correspond to vertices (*V*). Vertices connect when balls overlap, intertwined by edges/connections. Vertices are also weighted by the amount of points covered by it. These are visualised by discs, coloured based on a function $f{:}X \rightarrow R$.

In this study, a parameter size of ε = 0.3 was used as this provides a sufficient trade between granularity and less detail for visualisation - an approach utilised in similar studies where its findings had its robustness inspected (Dłotko et al., 2022; Rudkin et al., 2024) as well as relatively high external validity in the social science context (Dłotko, 2019). To validate this, this study conducted its own robustness check by; 1) attempting different ball radii, and 2) expanding the axis variables (see Appendix D). The purpose of the former is to explore the sensitivity of the BM analysis, while the latter attests to the reliability of applying TDABM when it involves variables with common methods vouched for by literature. Checking for these in assessing methodological choices are considered ideal practices as underscored by Dłotko et al. (2022).

4. Data

This study collected data from the 2021 Census[7]. As it was the most up-to-date data comprising the most accurate and reliable information, this dataset was prioritised as variables corresponding to the 2024 UK GE vote share. However, it is important to bear in mind that this data was collected online during COVID-19, where remote work and lesser economic activity was common – these dynamics have shifted since then. Due to the different political compositions and party systems (Rudkin et al., 2023), Scotland and Northern Ireland were excluded in this analysis, therefore only regions in England and Wales were included.

To observe how these variables affect voting behaviour, Census datasets were merged with the 2024 UK GE voting results data[8], downloaded as comma-separated values (CSV) (Cracknell & Baker, 2024) through its unique constituency code. This dataset also gives information on the electoral changes since the 2019 GE which is practical in considering how economic downturns like COVID-19 and Brexit may have shifted party polarisation overtime[9].

Prior to merging, datasets were cleansed and preprocessed on Excel, then loaded into Python where data transformation was done, by conversion of all units into floats and merging them together. Each variable was considered a single file/dataset. The rationale of converting data is for enhanced compatibility for analysis, while the merge is to enable a digestible analysis for the joint distributions. The analysis focused on the percentage of the

[7] Available at: [https://statistics.ukdataservice.ac.uk/organization/office-national-statistics-national-records-scotland-northern-ireland-statistics-and-research]
[8] UK GE24 data available at: [https://commonslibrary.parliament.uk/research-briefings/cbp-10009/]
[9] Refer to **Appendix A3** on exploratory data analysis comparing GE19 and GE24

dataset rather than its raw value, solely for the purpose of a universal scale within all datasets. Westminster Parliamentary constituency-level variables investigated for this analysis become axis variables in TDABM[10].

For the purpose of regression analysis, the merged dataset was further reduced, where the independent variables were particularly selected following literature on voting behaviour analysis. In ensuring the variables selected are also statistically consistent, feature selection was done to reveal the most predictive variables for voting patterns[11]. A criteria of validity for this study established that the use of variables with very high correlations above 0.7 were avoided as this may suggest redundancy. This is common practice in linear regression as demonstrated in literature (Rudkin et al., 2023; 2024). The reduced dataset included the following variables:

Variable type	Variable	Description (in %)	Abbreviation
Political	Outcome	Labour over Conservative vote shares in GE 24	'Lab-Con24P'
Socioeconomic	Education (qualification)	No qualification voters	'no_qualp'
	Deprivation (income level)	Voters deprived at all 4 levels	'dep4p'
	NS-SeC (employment)	Never worked/unemployed voters	'xworkedunemployedp'
Migration	Migrant indicator	Non-migrant voters	'nonmigp'
Demographic	Household by age	Single-family household voters under 50 years old	'singlefam16-49p'

Table 1: Reduced dataset for OLS & TDABM

[10] Refer to **Appendix A1 & Appendix A2** for full list of variables in merged dataset and summary statistics
[11] Appendix A4

% for 575 Constituencies	Variable	mean	std	min	max
2024 GE Vote	Valid Voters	59.84	7.04	39.99	75.21
	Conservative	24.73	10.12	0	53.3
	Labour	35.51	13.86	0	70.6
	Lib Dem	11.97	13.13	0	62.7
	Other	27.42	12.16	6.3	174.3
Qualifications (education)	Level 1	10.24	2.02	3	14.6
	Level 2	14.29	2.46	6.2	18.6
	Level 3	17.98	2.90	10.4	36.6
	Level 4	35.39	9.02	19.5	66.4
	None	19.21	4.88	4.9	34.2
	Other	2.89	0.49	1.4	5.4
Deprivation Indicator (income level)	None	48.41	5.94	32.1	62.7
	1	33.46	1.85	27.1	38.9
	2	14.20	3.11	7.5	22.4
	3	3.70	1.51	1.2	8.3
	4	0.23	0.13	0	0.8
NS-SeC (employment)	L1-L3	10.69	4.13	3.7	25.9
	L4-L6	16.36	2.71	8.2	21.9
	L7	9.39	1.37	4.9	13.3
	L8-L9	8.68	2.15	4	15.7
	L10-L11	4.44	1.03	1.7	7.8
	L12	9.32	1.96	4	14.1
	L13	9.82	2.99	3.1	19.9
	Never worked/Unemployed	6.81	2.38	3.1	16.3
	Full-time Student	6.10	4.13	2.9	38.7
Migrant Indicator	International	0.87	0.77	0.14	6.32
	Domestic	9.32	2.79	5.69	28.71
	Student	0.57	1.07	0.07	9.5
	Non-migrant	88.23	4.31	56.89	92.85
Household Composition (by age)	One-person (16-49)	3.71	1.57	1.5	13.3
	One-person (>50)	9.15	1.73	3.3	14.3
	Other/Mixed (16-49)	7.04	4.95	0	34
	Other/Mixed (>50)	3.17	0.99	0	8.9
	Single-family (<15)	16.88	1.97	9.8	22.2
	Single-family (16-49)	31.99	2.98	18	45.1
	Single-family (>50)	26.28	6.45	8.3	40.6

Table 2: Summary Statistics for Fully Merged Dataset

As the two main parties that continue to dominate British politics, only Labour and Conservative vote shares were observed for analysis, with Labour holding or gaining seats from Conservatives as being the outcome variable of this study. Observed in Table 2, there are varying ranges of variables. For example, the range of deprivation at all levels is relatively small compared to other variables like no deprivation. Due to this, all axis variables are normalised to [0, 1].

To visualise initial inferences of joint relationships between variables impacting voting behaviour, a correlation matrix through heatmaps and two-dimensional point clouds were mapped through scatterplots[12], where the points are coloured based on a binary outcome of distribution in which they belong. These two-dimensional linear relationships were then compared to the TDABM plots for the purpose of demonstrating how the latter unravel relationships that the scatterplots may miss in the following section.

[12] Summary statistics and heatmap of the reduced dataset for OLS & TDABM can be referred to in **Appendix B2 & Appendix B3**

5. Results

This study aimed to investigate the socioeconomic, migration, and demographic variables that impact voting behaviour in the 2024 UK GE by moving beyond traditional approaches and employing Topological Data Analysis. To compare and contrast the analytical approach of choice, an OLS regression model[13] was conducted for the outcome variable of Labour winning over Conservative, and the main predictors being voters of no qualification, of most deprivation, of unemployment, non-migrant status and in a single-family household of under 50 years old.

Correlation of selected variables revealed the strongest positive relationships between unemployment and the Labour vote, followed by voters who were the most deprived, younger in single-family households, have no qualification, and of international migrant status - respectively (Portes, 2023; Turner, 2022; Surridge, 2022). On the contrary, older populations in single-family households were most likely to not vote Labour, as well as being undeprived, of highest occupations, non-migrant status and highest qualification (Kulachai et al., 2023; McKnight & Cooper, 2022; McManus, 2023). Although this partially supports literature, mixed findings (see Simon et al., 2024; Bartels, 2016; Rudkin et al., 2023) indicate the heterogeneity of joint explanatory variables which will be further explored.

[13] OLS Regression Model (pre-robust) including its evaluations can be referred to in **Appendix B4**

5.1 OLS regression

This section is underpinned by the theoretical assumption of Anscombe (1973)'s quartet[14] who underscored how regression lines can appear the same despite differences in data structures. This study argues that model-based outcome variable approaches like OLS can mislook underlying trends in joint distributions, particularly when relationships between variables are non-linear. Typical in voting patterns, interactions between variables which influence this are far more complex and seldom linear due to varying voter profiles. Hence, a model like OLS cannot fully capture accurate relationships, especially its geographical variations vastly theorised in current political literature (Simon et al., 2024, Ford et al., 2021).

However, OLS maintains its strength of providing the significance level of each feature in observed clusters (Dłotko, 2019). Therefore, this advantage is used to investigate the significant relationships between sociodemographic independent variables and the voting outcomes, estimated as follows:

$$\textit{Lab-Con24P} = \alpha + \beta 1 \cdot \textit{no\_qualP} + \beta 2 \cdot \textit{dep4P} + \beta 3 \cdot \textit{xworkedunemployedP} + \beta 4 \cdot \textit{nonmigP} + \beta 5 \cdot \textit{singlefam16-49P} + \varphi$$

Its normality is diagnosed through assumption checks[15] to ensure that variables are statistically significant, even with literature attesting to it. Contrary to Simon et al. (2024), Kulachai (2023) and Otway and Rudkin (2024), the model found that predictors of no qualification and being deprived at all 4 levels are not statistically significant, but had large effects. Moreover, R-squared values suggested that only 51.1% of the variation in Labour

[14] In the visualised quartet, the mean, standard deviation, correlation and regression lines are all identical, yet when visualised, these plots show significantly differing distributions. See Appendix B1 for reference.

[15] Refer to **Appendix B5**

over Conservative vote shares could be explained by selected variables, which is moderate compared to voting studies indicating higher values (Evans & Tilly, 2017). This is understandable given that we hypothesise that relationships between explanatory variables and Labour votes are non-linear across geographies, justifying why linear regression models are insufficient. Insignificant variables are not removed to observe how TDABM can capture relationships where OLS cannot. To move forward with OLS statistical limitations, multicollinearity and non-normalities were addressed by applying robust standard errors to the model[16]. .

5.2 TDABM and point clouds in scatterplots

Comparable to scatterplots, TDABM is similar based on its given value from point clouds, but the latter enables more axis variables in a dataset given that data is sufficiently varied, and its colouration representing average outcomes visualises nuance in relationships - hence our use of normalised axis variables. Nevertheless, it should be emphasised that the variable set in TDABM for this section remains the same as OLS solely to permit consistent comparisons. Ideally, TDABM serves its advantage by including more axis variables to visualise relationships as it does not worry about correlation, unlike regression[17].

In the BallMapper analysis, each ball, or node, represents a collection of constituencies inclusive of the socioeconomic, migration, and demographic variables considered. The axis variables included the same variables for regression. The bigger its size, the greater the number of constituencies sharing the same characteristics as per axis variables

[16] This was chosen to adjust any standard errors by reflecting its authentic variability without impacting its goodness of fit, similarly demonstrated by Hobolt & Tilley (2016). Refer to **Appendix B6** for OLS Regression Model (Robust Standard Error)

[17] This inference was evaluated in the Robustness section in **Appendix D,** where further information on the study's robustness checks can be found.

- smaller outlier balls represent more unique characteristics. Its colour, ranging from warm to cool, shows the average range of values in these variables. Constituencies can lie within multiple balls, where edges indicate any constituency overlap. The seed and radius can be curated by choice – where the role of seed is to initialise the randomness of BallMapper positions to permit reproducibility when comparing, and the radius determines the granularity of plots as it adjusts the distances of balls from each other (Dłotko, 2019). Choice of the latter may amend TDABM structures, and consequently its interpretations, therefore robustness checks were done to ensure an appropriate radii that leverages between the noise of plots and potential detail reduction was chosen. It revealed that 1) ra

First, a BM graph with Labour winning over Conservative as the colouring variable was plotted (Figure 2). Notably, the graph showed clusters of constituencies with shared sociodemographic characteristics where the main nodes were concentrated among balls 0, 10, and 25. The mix of colour spectrums imply that although clusters may share other similar characteristics, voting behaviour varied. Ball 0 membership[18] being the biggest was shared across 298 constituencies – over half of that in England and Wales, with a majority of 193 Labour constituencies and 105 Conservative constituencies. Mapping these dispersed areas showed that constituencies sharing these variables were "middle England" (Moran, 2005), about 10% were Wales (Figure 3(b)). Nevertheless, this large cluster demonstrating homogeneity encompasses Red Wall, Blue Wall, and rural areas - some of which were regained as former strongholds lost (Kanagasooriam, 2019; Curtice, 2021).

Aligning with Otway & Rudkin (2024) who employed a similar TDA approach for the GE19 and Brexit vote, Labour seemed to perform best in isolated outlier balls (8, 18, 22),

[18] see **Appendix C** for full list of ball membership, its summary statistics and the mapping of constituencies

suggesting that these smaller constituencies have a unique mix of characteristics. These constituencies were Birmingham Ladywood, Leeds Central, and Sheffield Central respectively – known to be Labour strongholds in past elections (Bradbury, 2015). Indeed, Jennings and Stoker (2016, 2017, 2018) found stronger Labour support in urban centres and weaker in provincial areas.

On the contrary, the pattern of Conservative-dominant balls (26, 1, and 17) were more homogenous, with denser, interconnected, and bigger-sized balls (Otway & Rudkin, 2024). When visualising a map (Figure 3(c)), constituencies sharing similar behaviours were situated mostly in South (West & East) England and some part of Wales. These were established to be "Blue Walls" where Conservative votes were persistent (English, 2022; Curtice, 2021). However, mapping these areas in general show how Jenning and Stoker (2016)'s "two Englands" concept is reductive, as in reality, geographies interacting in multidimensions are more complex.

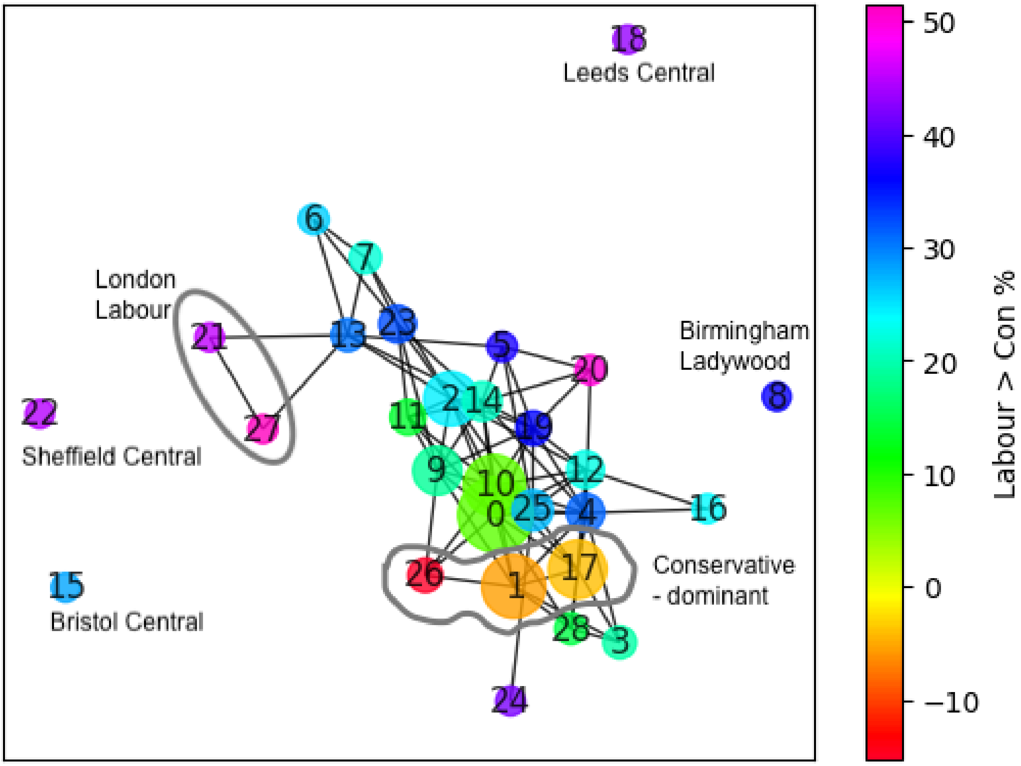
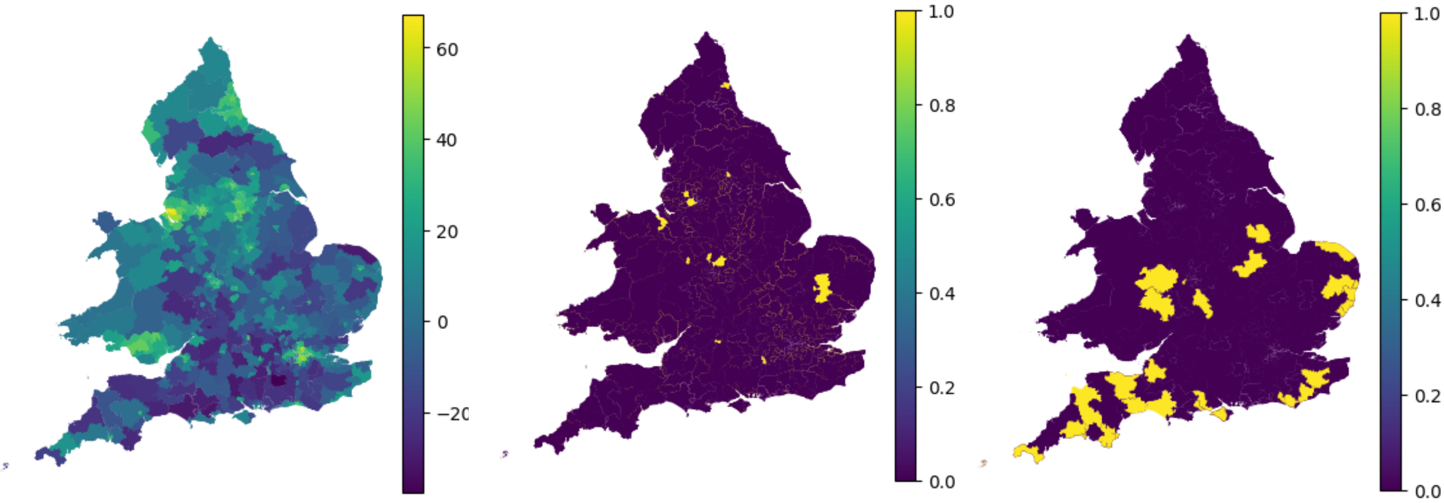


Figure 2: BallMapper Graph of Outcome Variable Labour Win



(a) (b) (c)

Figure 3: Mapping of Constituencies for (a) Labour %, (b) Ball 0, (c) Ball 26

Next, to evaluate how well the OLS model captured true voting behaviour based on the variables, the first TDABM graph was compared to this true value. A residuals graph was then constructed to observe the difference between the observed and true values (Figure 4). Comparing the outcomes, the model maintained both parties' strongholds, especially for Conservative, but Labour's performance was underestimated across many shared characteristic constituencies, especially when moving away from clusters. This suggests that the model failed to fully capture the actual strong Labour performance. However, in ball 8, Labour's strength was overestimated as Tory support was slightly higher than expected in this constituency. These discrepancies highlight the weakness of the OLS outcomes in capturing voter complexities, calling for a further model refinement by including more axis variables to fully comprehend the local sociodemographic context of constituencies' voting behaviour.

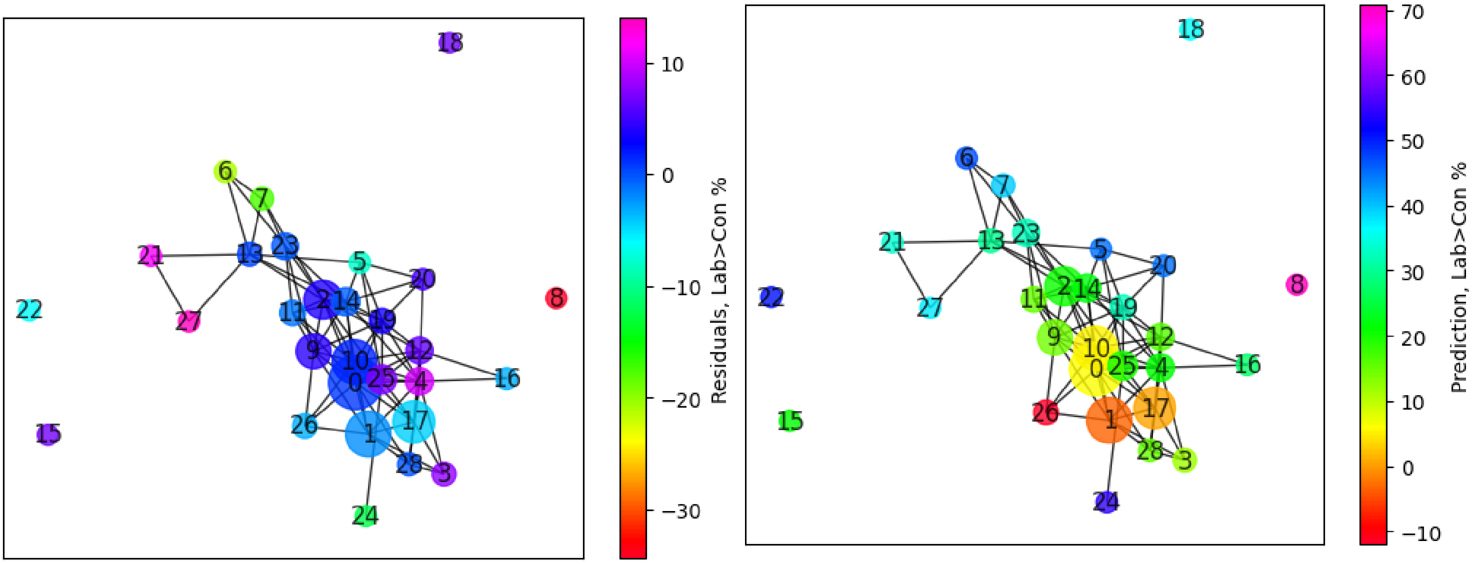


Figure 4: Residuals vs Prediction BM Graph

Thirdly, a visual comparison of scatterplots and TDABM graphs was done on the 5 predictor variables (Table 2) to show how moving beyond bivariate relationships offers insightful context (see Figure 5).

5.2.1 Education level

As seen in the scatterplots, the relationship suggested that as the percentage of people with no qualification increased, the share of Labour votes slightly increased as per our correlation (r = 0.31, p > 0.05). The TDABM agrees with this to an extent with joint distributions, but adds localised depth. For instance, in balls 9, 2, and 14 where constituencies' Labour wins were weak, the percentage of no qualification voters were higher than balls containing Labour strongholds (e.g. balls 21, 27, 20) which had lower no qualification voters – this granularity simply cannot be observed in the scatterplot. Indeed, Surridge (2020) observed how from 2010-2019, Labour-safe areas have been increasingly gaining voters of at least a degree qualification, diminishing the outdated label of Labour being the 'working-class' party when studies were of limited statistical inference.

5.2.2 Income level

Deprivation at all 4 levels was chosen as a predictor as opposed to other deprivation levels as the correlation showed appropriate values ($r < 0.7$), however OLS indicated non-statistical significance notwithstanding its positive relationship with Labour ($r = 0.49$, $p > 0.05$). Consequently, its extremely low variability resulted in an unclear linear trend observed in the Labour vote of the scatterplots. For a clearer structure, a demographic hue was added. The existing main cluster showed that Labour performed better amongst younger single-family household voters with lower income as Jennings & Stoker (2019) found.

The BallMapper plots showed clearer structure and confirmed this with added variables. Some areas were more homogenous within constituency groups in terms of deprivation, given other variables and political outcome. For example, lower deprivation at all levels were more consistent in constituencies of swing areas and Conservative-leaning (Dorling, 2016), but stronger Labour areas were more scattered (Jennings & Stoker, 2017; Rudkin et al., 2023). This confirms the general heterogeneity of Labour votes over Conservative given its joint distributions.

5.2.3 Employment

TDA balls grouping Conservative-leaning constituencies showed lower unemployment rates than Labour constituencies. Whilst the scatterplot successfully depicted this linear positive trend with Labour ($r = 0.61$, $p < 0.05$), TDA's smooth colour transitions in the joint distributions revealed how as the balls move further away from the main cluster, unemployment rates increase. This implies that areas outside 'middle England' constituted higher unemployment voters, nevertheless these areas were more tightly connected to each other as they were still homogeneously linked to areas with better unemployment, challenging the concept of "left-behindedness" rifting geographies based on sociodemographics.

5.2.4 Migration

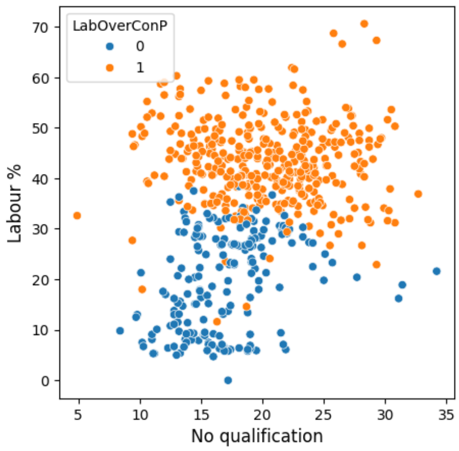
Understandably, voters who were non-migrants made up most of the population, observed in the higher end of scatterplots and the cooler tones in the TDABM, with a negative relationship with Labour wins ($r = -.26$, $p < 0.05$). Ball 20 had the lowest proportion of locals amongst interconnected nodes with very strong Labour influence - these were Liverpool and Manchester areas. This trend is similarly observed in other balls representing urban areas (balls 3, 4, 5, 28, 16) like Bath, Cambridge, Oxford and London cities, potentially attributed to higher student and young professionals migration (Curtice, 2017).

Despite non-migrant demographics dominating the population, Labour-leaning constituencies observed more diversity than Conservative constituencies which appeared more homogenous (Rudkin et al., 2023). Whilst the scatterplot might show this in spread out points, the TDABM's interconnected nodes showed how, notwithstanding colour intensity, constituencies were more closely connected in similar sociodemographic profiles when considering beyond migrant status alone.

5.2.5 Household composition and age demographic

There was a positive significant relationship between voters under 50 years old in single-family households and Labour votes (r = 0.35, $p < 0.05$), and the opposite for Conservative (Curtice, 2017). However, in the scatterplots, when adding a hue separating below/above average non-migrant areas to understand how demographic changes due to migration may affect voter choice, trends became unclear. These obscure visualisations may evidently risk misinterpretations when exploring beyond bivariate relationships, therefore TDABM provides this advantage of showing clearer groupings with multiple variables.

For example, the Conservative-dominant ball 26 showed significantly lower demographics of single-families under 50. On the contrary, ball 24 being a Labour stronghold had the highest young populations in single-family households. Yet, the interconnections of these two distinct balls indicate how joint distributions still exhibit shared behaviours, highlighting the importance of joint visualisations to capture electoral complexity.

<table>
<tr><th>Scatterplot</th><th>TDABM</th></tr>
<tr><td>


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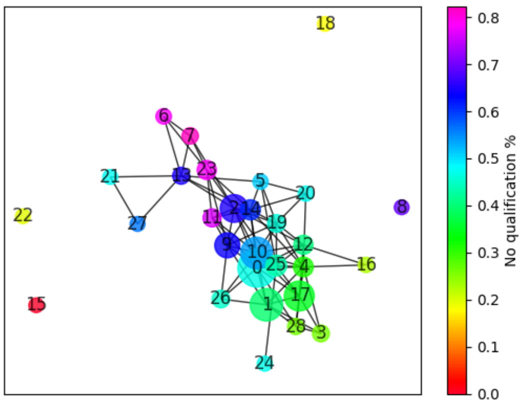

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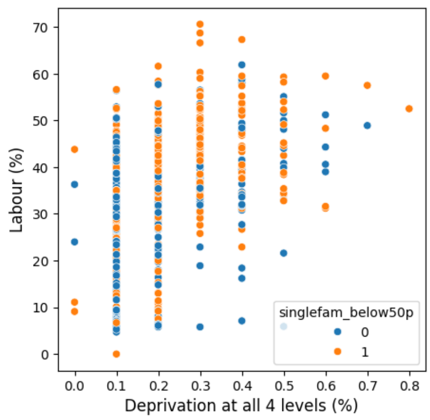
<tr><td colspan="2">A. No qualification (RQ1)</td></tr>
<tr><td>


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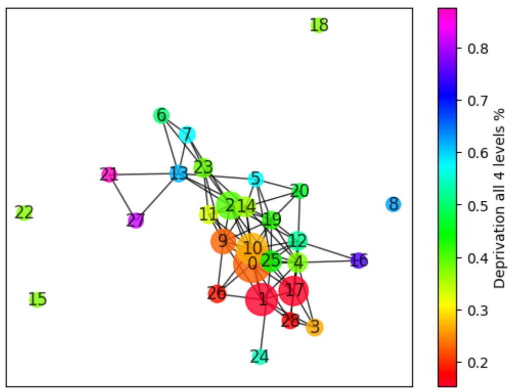

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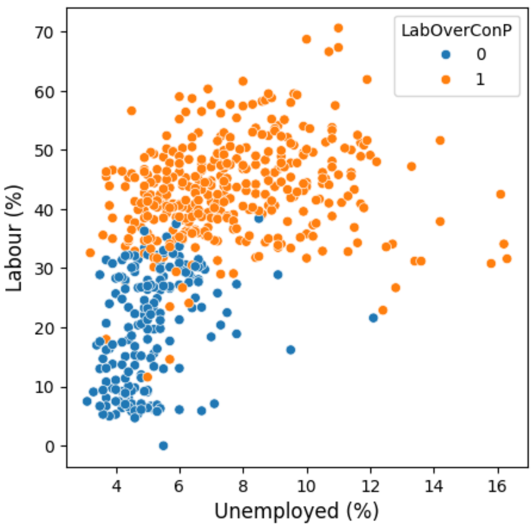
<tr><td colspan="2">B. Deprivation at all 4 levels (RQ1)</td></tr>
<tr><td>


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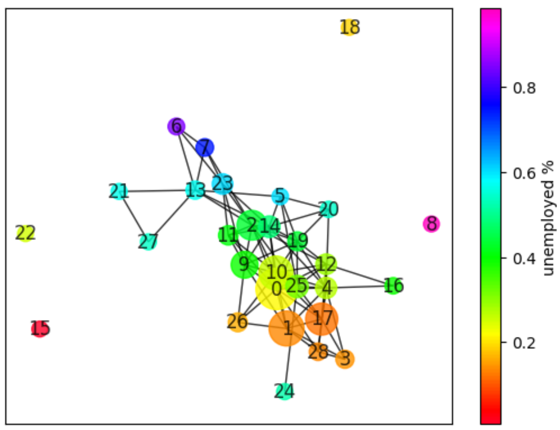

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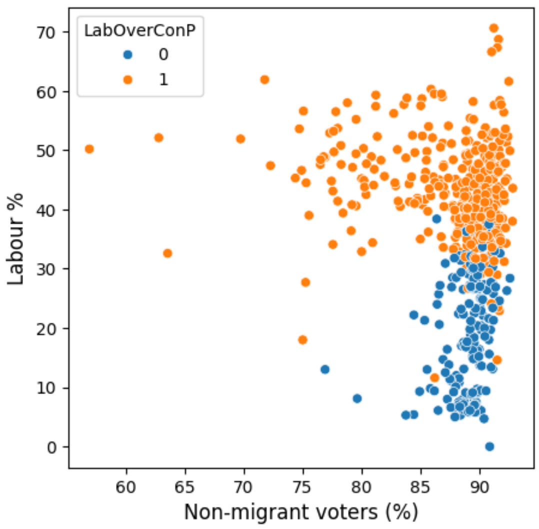
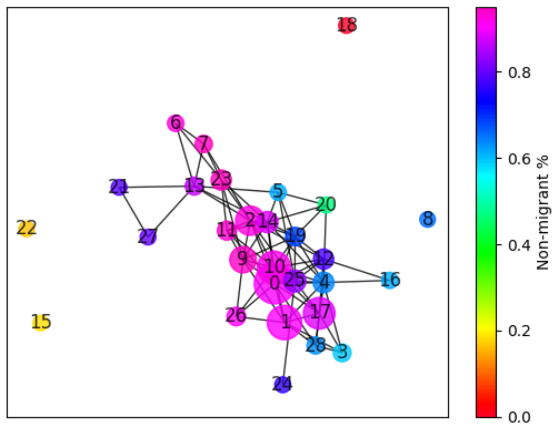
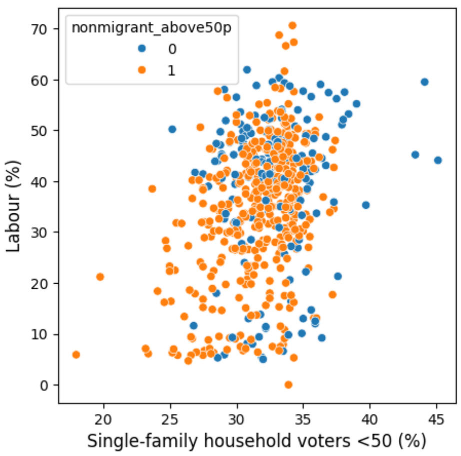
</table>

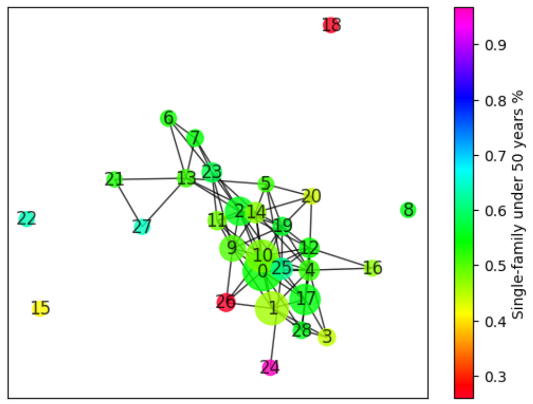
C. Unemployed/never worked (RQ1)





D. Non-migrant populations (RQ2)





E. Single-family household voters under 50 years old

Figure 5: Scatterplot vs TDABM plots for axis variables A-E

5.3 Outliers

Moving away from the focus on clusters sharing similar characteristics, apposite to TDABM is addressing the four detached balls behaving differently across all dimensions – these all contained single constituencies. Apart from the BM analysis allowing for the detection of unique character constituencies, its ability to provide global and local geographical insight is one that scatterplots do not allow, also being limited regression assumptions (e.g. homo/heteroscedasticity and normality).

- Ball 18 (*Leeds Central and Headingley)* and Ball 22 (*Sheffield Central*):

  Despite being distinct balls, both outliers which are Labour strongholds showed similar voting behaviours, with Sheffield Central showing generally warmer colour intensity than Leeds. This is potentially due to its different demographics – Leeds and Sheffield are known to be diverse and student cities (Johnston et al., 2018; Zhang, 2018), and the BM showing higher migration populations also suggest agreement to this trend. But whilst Leeds showed extremely low proportions of younger populations in single-family households, Sheffield had high proportions. Otherwise, both constituencies generally had low percentages of voters with no qualification and unemployment rates – interestingly different patterns from what the regression and the Labour-majority BM clusters tells us.

- Ball 8 (*Birmingham Ladywood)*

  Contrary to other outliers, Birmingham Ladywood had similar demographic profiles to the rest of the clusters, but observed extremely high socioeconomic proportions of no qualification voters, most deprived and unemployed (British

Election Study, 2019). As this was a constituency held by Labour in the last two elections, these profiles agreed with pocketbook voting, as Labour's social welfare policies is commensurate to the socioeconomic state of these voters (Evans & Tilley, 2017). Ethnically diverse populations in urban cities like Birmingham Ladywood also agreed with Johnston et al. (2018) hypothesising positive relationships with Labour.

- Ball 15: Bristol Central

Bristol Central is unique as it lost its Labour seat to Green who also had the highest vote share in the party in GE24 (Cracknell & Baker, 2024). This constituency consisted of high proportions of qualified and employed local populations, agreeing with Jennings & Stoker (2017) due to its young and progressive demographic. A common theme that remains is the fact that these two parties prioritised more liberal policies, reflecting on the social values young and highly-educated diverse voters emphasise (Surridge, 2020; Evans & Tilley, 2017). Bristol's case of Green party shift makes an essential study in understanding political realignment of left-wing parties as an alternative focus to "Red Wall".

6. Discussion

Although we previously addressed literature underlining the multivariates which linearly influence voting behaviour, such empirical models form limited relationships, potentially leaving out regional differences. Therefore, a complementary TDABM approach was taken to examine character groups at constituency-level, emphasising whether or not shared constituencies possess similar characteristics when migration, socioeconomic, and demographic variables adjoin. This moves away from the focus on investigating separate variable impacts on Labour/Conservative votes like in OLS - which we underscore as not the main aim of this study. Vast bodies of literature have already established this.

Whilst BM's memberships show local propensity to vote Conservative/Labour, we also do not establish causal relationships driving party choice. Categorising multidimensional distributions this way is simply an innovative approach to understanding voting behaviour due to three reasons; 1) the multifaceted information it provides, 2) as an initial exploration to further investigate joint relationships impacting voting choice given the ever-changing socioeconomic conditions across electorates, and 3) how local campaigns targeting particular sociodemographics may impact ball membership areas.

Evidently, Abreu & Öner (2020) and Otway & Rudkin (2024) have exhibited that multivariate influence at constituency-level in Brexit and UK GE19 votes were large. This paper therefore reproduces these approaches on GE24 amongst constituencies - categorising influences based on socioeconomic, demographic, and migrant contexts. In this section, we relate back to the three main research questions formulated based on the current body of literature to discuss what TDABM findings imply.

Overall, BM outcomes highlight a main theme – that Labour voters' sociodemographic patterns were more heterogeneous than Conservative voters who were more characteristically similar to each other. Although the main focus of this analysis was on the recent GE24 results, references to the previous elections GE19 are made to observe how voting behaviours may differ, given current data.

6.1 *Do socioeconomic variables predict constituency-level voting patterns?*

Commensurate with the theoretical frameworks (e.g. pocketbook, RCT) and empirical findings verifying this, the BallMapper analysis supports the idea that those socioeconomically vulnerable distinctly favoured Labour- which prioritise social welfare policies. This implies that voters' socioeconomic status do, indeed, predict constituency-level voting patterns given their demographic and migrant state. In the recent elections, Labour emphasised on policies that ensured economic stability, addressing income tax and VAT rates in response to economic events that disadvantaged these populations[19]. As unemployment was the second top national concern (Ipsos-MORI, 2022), employment provisions and protecting worker rights seemed to also be a top focus to help people get into jobs, with Labour introducing intentions to increase NHS staff pay for out-of-office hours, setting up a clean energy firm to hire 650,000 workers, and active teacher recruitment in England and Wales.

However, the mixed relationships from BM outliers when viewing joint relationships help explain deeper context as Lindner et al. (2020) suggested that in areas where educated

[19]All references of Labour manifestos discussed in this section can be accessed through: https://labour.org.uk/updates/stories/labour-manifesto-2024-sign-up/

voters are prevalent, left-wing parties tend to flourish more than in areas of working-class employment – apparent in Western democracy where voting behaviours have moved beyond political values (left vs right), prioritising social values (strongly associated with education) (Surridge, 2018). This shift underlines a dire need to explore reflective approaches on electoral studies.

Although policies from GE19 differed where Labour lost, their socioeconomic themes, and its relationship with voting behaviour maintained. The rising concern with the cost of living and unemployment recently could potentially manifest strong Labour preferences over parties with slightly different priorities like Conservative. Nevertheless, our TDABM findings highlight rural and urban contextual differences in the GE24 vote as the "left-behind" concept highlights, notwithstanding Labour strength in these Labour-winning areas. Notably, Labour strongholds like Leeds, Sheffield and London constituencies (seen away from clusters and outlying balls) showed higher qualifications, and more diverse populations as opposed to rural Labour constituencies with higher socioeconomically vulnerable profiles (mostly non-migrants deprived at all levels, had no education and were unemployed). This nuance supports rising literature on the urban-rural divide (Goodwin & Heath, 2016), reflecting on how urbanised areas continue to support progressive policies while rural populations take advantage of them. Labour has also growingly become the party of the 'educated' even pre-Brexit(Surridge, 2020), moving away from its traditional base of poor qualification voters. Indeed, socioeconomic statuses and constituency distinctions are crucial in understanding voting choice.

6.2 *To what extent do migration patterns post-Brexit impact the political landscape?*

Whilst this study sought migrant status data from the 2021 Census, migrant populations in England and Wales since then continue to rise, notwithstanding post-Brexit and the past Conservative rule intending to limit it (ONS, 2024). The 2019 GE noticed that lower immigration areas favoured Conservatives (attributed to 'left-behindedness'), as this was linked to the party's promotion of anti-immigration sentiments prior to and throughout Brexit (Goodwin & Heath, 2016). This study's observation of the homogeneity in Conservative clusters when considering migrant populations are consistent with this in GE24, despite Labour's win post-Brexit. It also implied migration still being a divisive concern in shaping political scenes, as Labour held dominance in more diverse urban areas, regardless of political value.

Additionally, Surridge (2020) underscored how Labour's Red Wall areas increasingly lost prior to the GE24 cannot be attributed by Brexit alone, further highlighting the prolonging political realignment. Whilst this study agrees with Surridge (2020) that Brexit was an expedient factor rather than a cause of Conservative wins, GE24 evidently showed massive Labour regains in these Red Wall areas where our TDABM agrees. This implies that the political landscape post-Brexit does not necessarily hold this concept of realignment, rather likely due to poor Conservative performance during its incumbency as Fieldhouse & Bailey (2023) agrees.

For instance, whilst the 2024 Conservative manifestos focused on addressing immigrants in the UK (e.g. sending asylum seekers to Rwanda and Visa quota)[20], Labour on

[20] All references of Conservative manifestos discussed in this section can be accessed through: https://www.conservatives.com/manifesto

the contrary were against deporting asylum seekers and provided alternatives to curb illegal immigration. Conservatives also focused on managing employed taxpayers' spendings (e.g. corporation taxes), whilst priority was taken by Labour on local workers' job security, by ending reliance on expatriates in healthcare and construction. These policy distinctions which address both voter socioeconomic and migration intersections showed that, regardless of Brexit or political realignment, these factors played a crucial role in voting choice where this study aimed to investigate.

6.3 *Can demographic changes predict shifts in voting behaviour from migration trends?*

With the politicisation of migration trends in mind, polarisation in voting behaviour of diverse youth into urban/University cities like Bristol Central could be a testament to how demographic shifts can realign regional politics, observed in our study. Referring to current findings, whilst Surridge (2020) similarly studied the diversity of increasing migration rates in cosmopolitan areas, reduced demographic activity in rural constituencies resisted Conservative support - even in prior elections (Ford & Goodwin, 2014). The study's joint distributions, especially in the BM plot for constituencies in East London/Bristol/Leeds (strong Labour outliers), highlight the role of how these factors interplay to drive this change.

Due to the cost of living crisis rising, household composition is imperative in understanding the policy choices that may drive voter behaviour. Labour pledged to increase childcare in England through initiatives like free school breakfasts and more nurseries in GE24, attracting family households as it benefits them. To favour one-person households, Labour also prioritised first-time property buyers to purchase over investors, and introduced mortgage schemes. This appeals to voters potentially suggesting an impact of voting behaviour, underlining the need for upcoming political strategies to mitigate complex sociodemographic interactions of the population.

7. Conclusion

This study sought to comprehend how dynamic socioeconomic, demographic, and migration contexts impact voting behaviours at constituency-level in the 2024 UK GE, as the knowledge of driving factors remains limited due to its recency. By complementing with OLS regression, this study implemented TDABM, highlighting its advantages as opposed to standard regression models widely used in the current political field. Findings of joint distributions agree with literature, offering deeper contexts of regional nuances that are limited in current approaches. The extent to which these may affect voting choice was discussed, where factors like voters' socioeconomic classification, qualification and employment state heavily influenced party preference especially in rural areas seen in ball clusters and urban areas in outliers. Additionally, migration rates in areas were also examined and showed heterogeneity in Labour voters in both economically advantaged and disadvantaged areas, more so than Conservative who maintained favouring anti-immigration under the pretense of preserving national sovereignty.

As BallMapper does not establish causality, visualisations that align with statistical inferences were the motivation of choice in methods studying multivariates. Regardless of limited or expanded variables used in the BM, our unique application of TDA provides a key contribution of richer understanding on how similar sociodemographic and migration profiles interact in geographical spaces.

Since the last GE, many economic events driving policy changes, whether successfully implemented or not, have occurred to address the country's socioeconomic state - steering voter choice and consequently electoral wins. As these variables become important factors in voter preferences, joint distribution findings of this study therefore argue on the importance that party manifestos and policies play, practically implying that political parties should attempt to bridge the divide between urban-rural bases through mitigating economic insecurity accordingly. This is to ensure voters who are 'left-behind' are not marginalised, in hopes that future trends in upcoming elections are more homogenous regardless of party choice, indicative of economic prosperity.

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